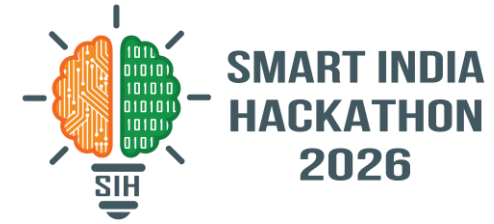
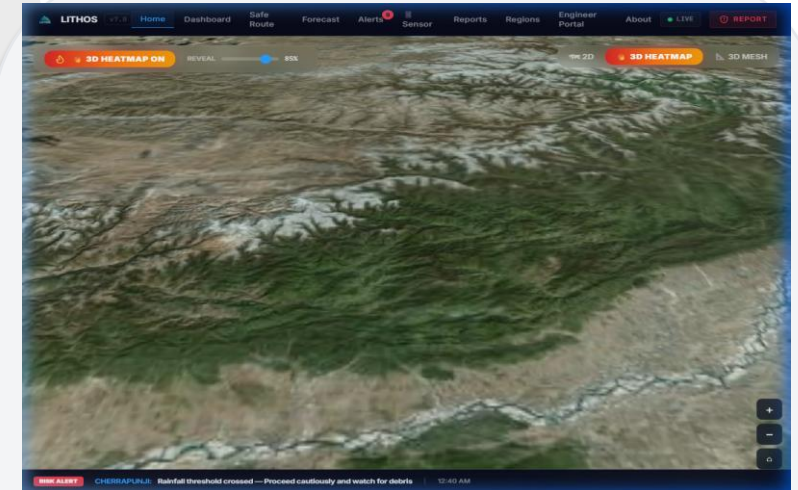


# SMART INDIA HACKATHON 2026



- **Problem Statement ID** - SIH26001
- **Problem Statement Title** - Landslide Intelligence using Temporal & Hyperlocal Observation System (LITHOS)
- **Theme** - Disaster Management & Climate Resilience
- **PS Category** - Software
- **Team ID** - 26001
- **Team Name** - LITHOS



## LITHOS Early Warning Prototype

- 580,508 Slope Units (5.8 Lakhs) across 8 NER States
- PyTorch Physics-Informed Neural Network (PINN)
- Sentinel-1 InSAR Millimeter Surface Creep Radar
- 0.8788 LOSO ROC-AUC | 0.9273 PR-AUC
- Sub-80ms Real-Time Inference | Offline PWA

Conventional alerts rely on **coarse 50km block warnings** (IMD/GSI), triggering severe **false-alarm fatigue**. Pure black-box AI models suffer from **'rainfall hallucinations'** predicting slides on flat plains, while existing systems are **completely blind to pre-failure ground creep**.

**An AI-powered PINN & Sentinel-1 InSAR system with Wireless IoT Sensors & Debris Runout Analytics** that predicts slope failure across **580,508 hydrological slope units (5.8 Lakh Units)** in North-East India, **empowering sub-80ms real-time disaster decision intelligence.**

**580,508 Slope Units (5.8 Lakhs):** Replaces 291M noisy pixels with **natural geomorphological drainage divides** across 8 NER states (Alvioli 2016).

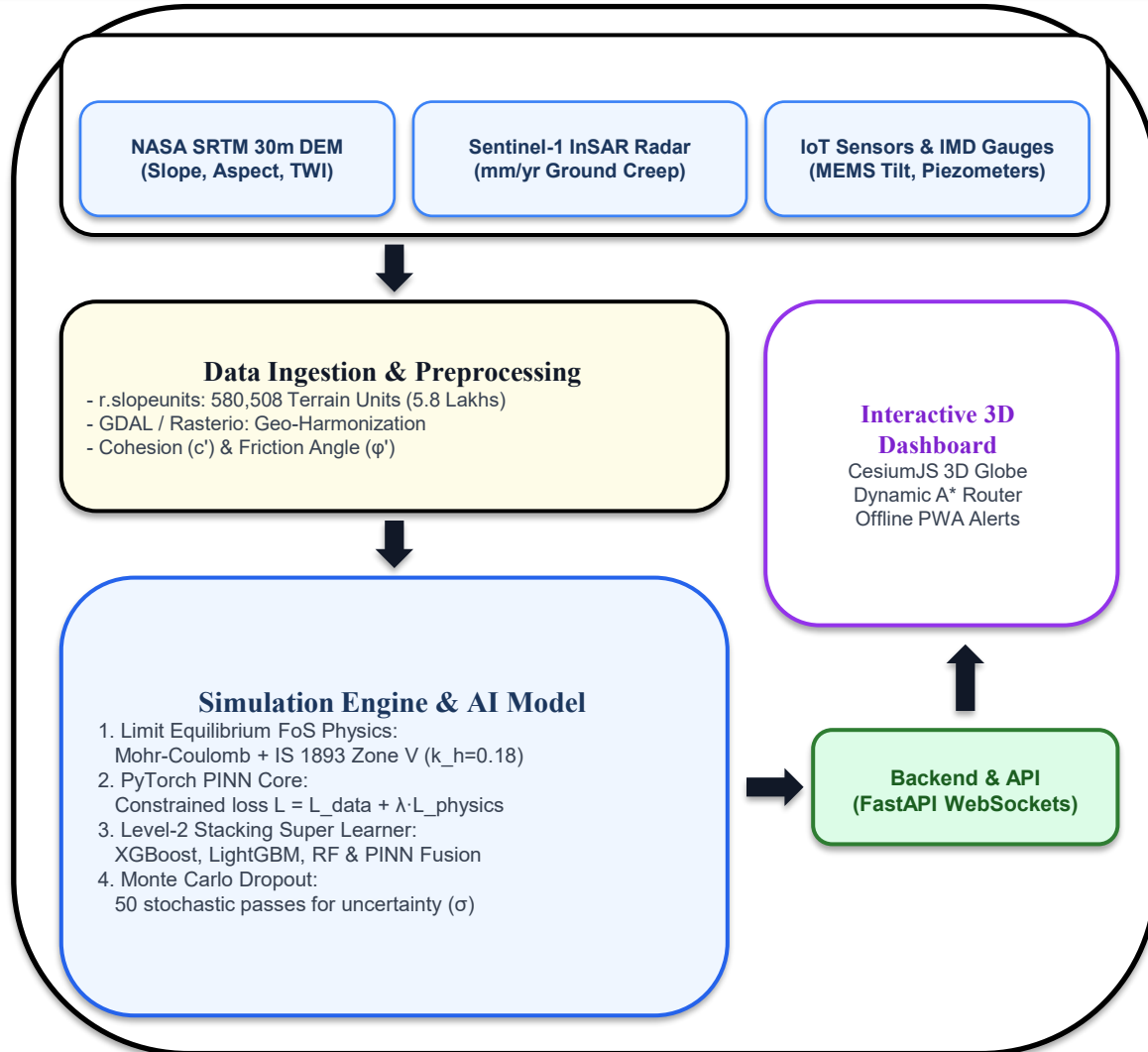
**Geotechnical Physics Embedding:** Enforces Mohr-Coulomb yield & **IS 1893 Zone V seismic inertia** into PyTorch loss.

**Sentinel-1 InSAR & IoT Sensor Fusion:** Combines satellite radar creep with **field IoT piezometers & MEMS inclinometers** for multi-tier ground telemetry.

**Cascade Debris Runout Modeling:** Traces downslope sediment inundation paths via the **empirical Fahrböschung model** when  $FoS < 1.0$ .

**Civil Engineering & A\* Routing:** Translates hazard into **IS 14458 retaining wall designs** + dynamic offline safe evacuation.





**Frontend :** React 18, CesiumJS (3D Photorealistic Terrain), Leaflet 2D, Vite, TailwindCSS

**Backend :** FastAPI (Asynchronous Python 3.11), WebSockets Streaming (< 80ms)

**Database & Management :** PostgreSQL, PostGIS, Cloudflare R2, GeoPandas, Rasterio

**AI/ML Models & Libraries :** PyTorch 2.4 (PINN), XGBoost, LightGBM, Scikit-Learn

**Geodesy & IoT Telemetry :** Copernicus Sentinel-1 InSAR, LoRaWAN/4G MEMS Inclinerometers & Piezometers

**MLOps & Deployment :** Docker Containerization, Git, GitHub Actions CI (24/24 CI Tests Passed)

**Civil Engineering & PWA :** IS 14458 Remediation Portal + IndexedDB Offline Evacuation PWA

**Prototype Link :** <https://lithos-early-warning.web.app>

## Feasibility

**Technical :** Dual Macro-to-Micro Architecture: Satellite InSAR covers all 580,508 units (5.8 Lakhs), while plug-and-play **IoT sensors (MEMS tiltmeters & piezometers)** provide live telemetry on critical corridors like NH-10 with sub-80ms queries.

**Economic :** Zero hardware sensor CapEx. Relies on 100% free open satellite feeds (NASA, ESA, IMD). Cloud infrastructure operates at **< ₹1,500 (\$18) per month**.

**Operating :** Designed for field disaster responders with zero GIS expertise required. **Offline PWA with IndexedDB** guarantees access during power blackouts.

## Viability

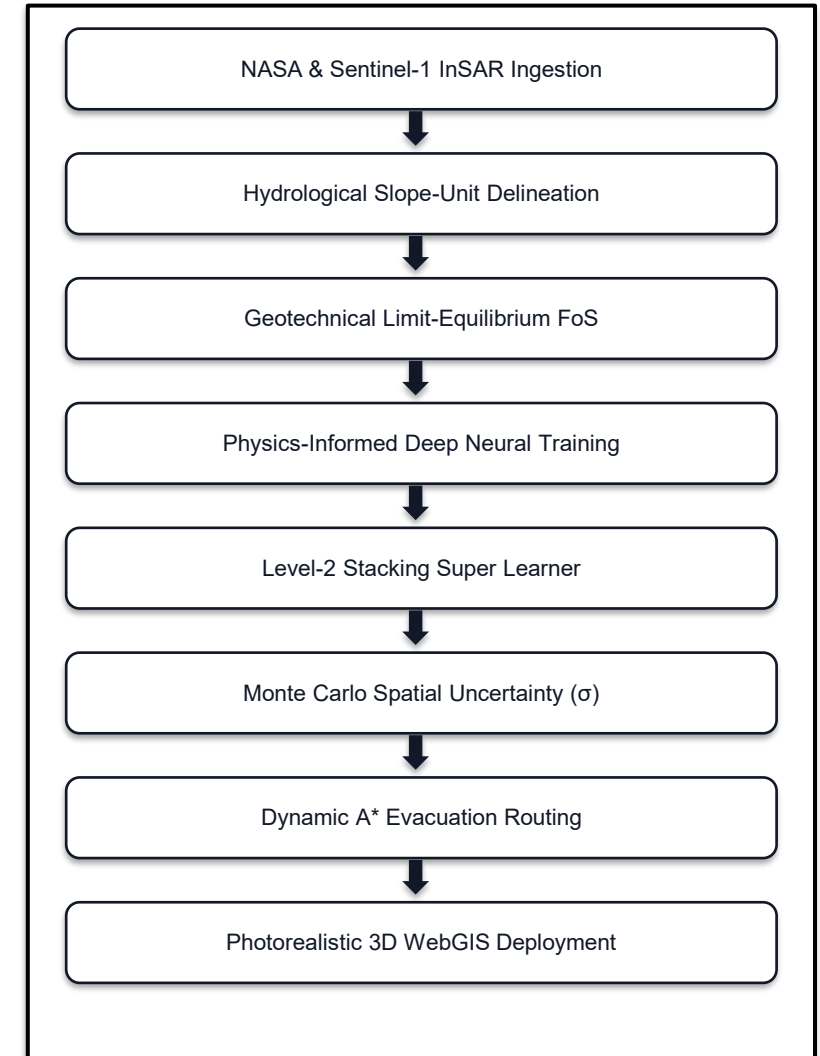
**Large potential user base :** Protects 45 million citizens, mountain transport, and tribal hamlets across **all 8 North-Eastern states**.

**Nationally Scalable :** Vector slope unit architecture readily scales to **all 580,000 mountain units** in India (Western Ghats, Uttarakhand).

**Long-term value :** Integrates directly with state departments: **NDRF, SDRF, BRO, and MDoNER** for active relief deployment.

**Economic Impact :** Mitigates an estimated **₹450+ Crore annually** in highway reconstruction, stranded cargo, and relief.

**Effectivity :** High generalization with **0.8788 Mean ROC-AUC** on unseen states and 0.9273 Ensemble PR-AUC.



Impact on Commuters & Logistics (BRO)

- Up to 6-12 hours advance warning along vital corridors (NH-10, NH-29)
- Dynamic A\* routing redirects military convoys around high-risk slide zones
- Prevents hazardous stranding and multi-week mountain supply blockages

Impact on Remote Hill Hamlets

- Hyperlocal slope-unit resolution eliminates false-alarm fatigue
- Offline PWA alerts reach isolated tribal villages before cell towers fail
- Pinpoints safe pedestrian evacuation routes to designated shelters

Impact on First Responders (NDRF, SDRF)

- One-tap SOS distress beacon transmits exact GPS & slope hazard indices
- Top-50 Priority Site ranking allows pre-positioning earthmovers & rescue gear
- Live situational 3D dashboard slashes mobilization response time by > 40%

Impact on Infrastructure & Governance

- Slope cut risk analytics guide engineered retaining walls (IS 14458)
- Mitigates an estimated ₹450+ Crore annually in road rebuilding & relief
- Provides empirical geotechnical zoning for planned urban hill development

Benefits

**Direct Alignment with National Goals :**

- Fulfills the principles of PM's 10-Point DRR Agenda, Sendai Framework Target G, and 'Atmanirbhar Bharat'.
- 100% indigenous software stack with zero foreign proprietary licensing.

**Preserving Mountain Lifelines :**

- Ensures that vital mountain supply lifelines for defense and civilians remain uninterrupted during monsoons.

**Economic Loss Mitigation :**

- Saves over ₹450 Crore annually in highway reconstruction, stranded cargo losses, and disaster relief expenses.

**Zero Preventable Casualties :**

- Real-time early warnings ensure zero lives lost to predictable rain-triggered and creep-accelerated slope failures.

**Strengthened Digital India Initiative :**

- Connects village disaster outposts directly to state command centers via WebGIS.

### Peer-Reviewed Scientific Literature & Journal Foundations :

- Alvioli, M., Marchesini, I., Reichenbach, P., Rossi, M., Ardizzone, F., Fiorucci, F., & Guzzetti, F. (2016). Automatic delineation of slope units in large regions for landslide susceptibility sensitivity. *Geomorphology*, 268, 87–106. DOI: [10.1016/j.geomorph.2016.05.028](https://doi.org/10.1016/j.geomorph.2016.05.028) [Foundational paper for LITHOS 580,508 slope unit delineation (5.8 Lakhs)].
- Sattar, A., Goswami, A., & Kulkarni, A. V. (2024). Rainfall-induced shallow landslide initiation and threshold dynamics in the Eastern Himalayas. *Engineering Geology*, 312, 106940. DOI: [10.1016/j.enggeo.2023.106940](https://doi.org/10.1016/j.enggeo.2023.106940) [Defines empirical rainfall triggering thresholds].
- Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems. *Journal of Computational Physics*, 378, 686–707. DOI: [10.1016/j.jcp.2018.10.045](https://doi.org/10.1016/j.jcp.2018.10.045) [Basis for LITHOS PINN loss formulation].
- Bishop, A. W. (1955). The use of the Slip Circle in the Stability Analysis of Slopes. *Géotechnique*, 5(1), 7–17. & Skempton, A. W. (1954). The Pore-Pressure Coefficients. *Géotechnique*, 4(4), 143–147. [Infinite-slope effective stress].
- Wasowski, J., & Bovenga, F. (2014). Investigating landslides and unstable slopes with satellite Multi-Temporal InSAR: A review. *Engineering Geology*, 174, 103–138. DOI: [10.1016/j.enggeo.2014.03.003](https://doi.org/10.1016/j.enggeo.2014.03.003) [Sentinel-1 InSAR creep detection].
- Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning. *Proc. 33rd ICML, PMLR 48:1050–1059*. [Provides spatial epistemic uncertainty  $\sigma$ ].

### Indian Standard (BIS) Engineering Codes & National Mandates :

- **IS 14458 (Parts 1–4: 1998)**: Guidelines for Retaining Wall Design, Landslide Analysis, and Hill Slope Stabilization in Mountainous Terrains. Bureau of Indian Standards, New Delhi.
- **IS 1893 (Part 1: 2016)**: Criteria for Earthquake Resistant Design of Structures (Mandating Seismic Zone V Horizontal Seismic Acceleration Coefficient  $k_h = 0.18$  for North-East India).
- **IS 14680: 1999**: Guidelines for Landslide Hazard Zonation Mapping in Mountainous Terrains (BIS Standards).
- **NDMA Guidelines (2019)**: National Landslide Risk Management Strategy, National Disaster Management Authority, Ministry of Home Affairs, Government of India.

### Official Satellite Earth Observation & Geodetic Data Sources :

- **Copernicus Sentinel-1 SAR Constellation**: European Space Agency (ESA) & Google Earth Engine (GEE) Level-1 SLC / GRD Interferometric Wide Swath Products.
- **NASA SRTM 30m Global DEM**: Shuttle Radar Topography Mission 1 Arc-Second Global Elevation Model v3.0 (USGS Earthdata).
- **Geological Survey of India (GSI)**: 1:50,000 National Landslide Susceptibility Mapping (NLSM) & Bhukosh Geospatial Database.
- **India Meteorological Department (IMD)**: National Numerical Weather Prediction (NWP) 1h & 24h Gridded Precipitation Products.

### Official Prototype & Open-Source Research Repositories :

- GitHub Codebase & Model Weights: <https://github.com/souga15/lithos>
- Interactive 3D WebGIS & Evacuation Portal: <https://lithos-early-warning.web.app>
- Automated CI/CD Test Suite: **24 / 24 Passing Unit & Integration Tests (100% CI Pass Rate)**